

PROJECT STATE — Kamai Backend OMS

Single source of truth for implementation progress. Updated after every completed action.

Overall Progress

Metric	Value
Total Actions	TBD
Completed Actions	8
Remaining Actions	TBD
Current Action	Action 10 — awaiting sequence diagram
Total Actions	24
Completed Actions	24
Remaining Actions	0
Current Action	MVP Complete

Completed Actions

- ✔ **Foundation Setup** — Project scaffold, Docker, CI/CD, tooling, shared infrastructure
- ✔ **Authentication** — Firebase Phone Auth Login (**POST** `/api/auth/firebase/login`), Logout (**POST** `/api/auth/logout`), Baker model, RefreshToken model, session issuance
- ✔ **Action 3 — Summary Dashboard** — **GET** `/api/dashboard/summary`, **Order** model, JWT auth middleware

- **✓ Action 4 — Create New Order** — **POST** `/api/orders`, **Customer** model, Prisma transactions, Caching/Audit stubs
- **✓ Action 5 — Order History** — **GET** `/api/orders`, pagination, multi-sort, dynamic filters, compound indexes
- **✓ Action 6 — View Order Details** — **GET** `/api/orders/:orderNumber`, unified detailed DTO retrieval
- **✓ Action 7 — Update Order Status** — **PATCH** `/api/orders/:orderNumber/status`, state machine validation
- **✓ Action 8 — Record Balance Payment** — **PATCH** `/api/orders/:orderNumber/payment`, finance ledger, **PaymentStatus**
- **✓ Action 9 — Edit Order** — **PUT** `/api/orders/:orderNumber`, update order details and recalculate financials
- **✓ Action 10 — Cancel / Archive Order** — **DELETE** `/api/orders/:orderNumber`, soft delete logic, terminal state validation
- **✓ Action 11 — Customer Upsert** — Automatic CRM creation/update during order placement, LTV calculation
- **✓ Action 12 — Customer Directory** — **GET** `/api/customers`, sort/search CRM list, outstanding balance calculation
- **✓ Action 13 — Customer Profile** — **GET** `/api/customers/:customerId`, full profile with paginated order history
- **✓ Action 14 — Update Customer Profile** — **PUT** `/api/customers/:customerId`, edit CRM profile, deduplicate phone numbers
- **✓ Action 15 — Calendar View** — **GET** `/api/dashboard/calendar`, aggregates orders by delivery date for month/week views
- **✓ Action 16 — Investment / Expense Ledger** — **POST, GET, DELETE** `/api/investments` for tracking raw material costs and net profit calculation
- **✓ Action 17 — Billing & Subscription** — **GET** `/api/billing/status`, **POST** `/api/billing/create-subscription`, Razorpay Gateway integration
- **✓ Action 18 — Razorpay Webhook Processing** — **POST** `/api/webhooks/razorpay`, idempotent subscription event processing and **BillingHistory** ledger
- **✓ Action 19 — Manage UPI Settings** — **PUT** `/api/baker/upi-settings`, Baker profile extension with generic **PaymentSettingsService**
- **✓ Action 20 — Upload Business Assets** — **POST** `/api/uploads/signed-url`, **POST** `/api/uploads/confirm`, Supabase Storage integration with short-lived signed URLs
- **✓ Action 21 — WhatsApp Receipt & Payment Reminder Generation** — **POST** `/api/notifications/whatsapp`, deep link generation with separated

WhatsAppTemplateEngine, MessageFormatter, and DeepLinkGenerator

-  **Action 22 — Baker Profile & Settings** — `GET /api/baker/profile`, structured profile reading, mapped Settings DTO with dynamic storage URL resolution
-  **Action 23 — Support & Help Desk** — `POST /api/support/chat`, WhatsApp support deep link generation with `SupportMessageFormatter`, version detection, and `SUPPORT_NOT_CONFIGURED` 503 error handling
-  **Action 24 — Logout** — `POST /api/auth/logout`, per-device refresh session revocation, auth cookie clearance, and `USER_LOGGED_OUT` audit logging



Pending Actions

-  Action 10 — awaiting sequence diagram



Modules Completed

Module	Status	Notes
Foundation / Scaffold	 Done	package.json, tsconfig, Docker, plugins, error handling
Auth	 Done	Firebase token verify, baker create/login, JWT, refresh token, middleware
Baker	 Pending	
Orders	 Done	<code>Order</code> creation, history, details, status lifecycle API, payment status
Customers	 Done	Customer Directory (<code>GET /api/customers</code>), Profile (<code>GET /api/customers/:customerId</code>), Profile Update (<code>PUT /api/customers/:customerId</code>), Upsert logic
Dashboard	 Done	<code>GET /api/dashboard/summary</code> metrics API, <code>GET /api/dashboard/calendar</code>
Finance	 Done	<code>PaymentLedger</code> recording, Investment / Expense Ledger (<code>/api/investments</code>)

Module	Status	Notes
Billing & Subscription	 Done	GET /api/billing/status, POST /api/billing/create-subscription, Webhooks (POST /api/webhooks/razorpay)
Settings	 Done	Manage UPI Settings (PUT /api/baker/upi-settings), Uploads (POST /api/uploads/signed-url), Baker Profile (GET /api/baker/profile)
Calendar	 Pending	
Investments	 Pending	
Support	 Done	Support & Help Desk (POST /api/support/chat)
Notifications	 Done	WhatsApp Receipt & Reminder Generation (POST /api/notifications/whatsapp)
Uploads	 Pending	



Database Status

Item	Value
ORM	Prisma 5.x
Provider	PostgreSQL (Supabase)
Schema Version	v0.4.0
Latest Migration	20260725203945_add_order_history_indexes
Tables Created	4 (Baker, RefreshToken, Order, Customer)

Tables

Table	Key Fields
Baker	<code>id</code> (PK), <code>firebaseUid</code> (unique), <code>phoneNumber</code> (indexed), <code>status</code> , <code>subscriptionStatus</code>
RefreshToken	<code>id</code> (PK = sessionId), <code>tokenHash</code> (unique, SHA-256), <code>bakerId</code> (FK), <code>expiresAt</code> , <code>revokedAt</code>
Customer	<code>id</code> (PK), <code>bakerId</code> (FK), <code>phone</code> (unique with bakerId), <code>name</code>
Order	<code>id</code> (PK), <code>orderNumber</code> (unique), <code>bakerId</code> (FK), <code>customerId</code> (FK), <code>deliveryDate</code> , <code>status</code> , <code>totalPrice</code> , <code>advancePaid</code> , <code>balanceDue</code> , <code>deletedAt</code>



Environment Variables

Variable	Status	Required By
<code>DATABASE_URL</code>	✓ Configured	Supabase pooler (port 6543, pgbouncer=true)
<code>DIRECT_URL</code>	✓ Configured	Supabase direct (port 5432, for migrations)
<code>SUPABASE_URL</code>	✗ Pending	Uploads, Storage
<code>SUPABASE_SERVICE_ROLE_KEY</code>	✗ Pending	Uploads, Storage
<code>SUPABASE_ANON_KEY</code>	✗ Pending	Optional
<code>SUPABASE_STORAGE_BUCKET</code>	✗ Pending	Uploads
<code>JWT_SECRET</code>	✓ Configured	Auth (80-char hex)
<code>JWT_REFRESH_SECRET</code>	✓ Configured	Auth (80-char hex)
<code>JWT_ACCESS_EXPIRES_IN</code>	✓ Configured	Auth (default: 15m)
<code>JWT_REFRESH_EXPIRES_IN</code>	✓ Configured	Auth (default: 7d)
<code>COOKIE_SECRET</code>	✓ Configured	Auth (80-char hex)

Variable	Status	Required By
REDIS_URL	✅ Configured	Local Docker Redis (localhost:6379)
RAZORPAY_KEY_ID	❌ Pending	Payments
RAZORPAY_SECRET	❌ Pending	Payments
RAZORPAY_WEBHOOK_SECRET	❌ Pending	Payments
FIREBASE_PROJECT_ID	✅ Required	Firebase Admin SDK (Auth)
FIREBASE_CLIENT_EMAIL	✅ Required	Firebase Admin SDK (Auth)
FIREBASE_PRIVATE_KEY	✅ Required	Firebase Admin SDK (Auth)
TWILIO_ACCOUNT_SID	❌ Pending	WhatsApp Notifications
TWILIO_AUTH_TOKEN	❌ Pending	WhatsApp Notifications
TWILIO_WHATSAPP_FROM	❌ Pending	WhatsApp Notifications
RESEND_API_KEY	❌ Pending	Email
SENTRY_DSN	❌ Pending	Monitoring
CORS_ORIGIN	❌ Pending	Foundation
PORT	❌ Pending	Foundation (default: 3001)
NODE_ENV	❌ Pending	Foundation
LOG_LEVEL	❌ Pending	Foundation (default: info)



Third-Party Integrations

Service	Status	Notes
Supabase (Database)	✅ Configured	Connected & schema validated
Supabase (Storage)	⌚ Pending	Bucket name not yet provided
Redis	✅ Configured	Local Docker (kamai-redis, port 6379)
BullMQ	⌚ Pending	Redis ready, BullMQ not wired yet

Service	Status	Notes
Razorpay	 Pending	Keys not yet provided
Firebase Auth	 Integrated	Firebase Admin SDK v14 — POST /api/auth/firebase/login
Twilio / WhatsApp	 Pending	SID/token not yet provided (notifications only — NOT used for OTP)
Resend (Email)	 Pending	API key not yet provided
Sentry	 Pending	DSN not yet provided

API Progress

Method	Endpoint	Module	Status
GET	/health	System	 Done
GET	/docs	Swagger	 Done
POST	/api/auth/firebase/login	Auth	 Done
GET	/api/dashboard/summary	Dashboard	 Done
POST	/api/orders	Orders	 Done
GET	/api/orders	Orders	 Done
GET	/api/orders/:orderNumber	Orders	 Done
PUT	/api/orders/:orderNumber	Orders	 Done
PATCH	/api/orders/:orderNumber/status	Orders	 Done
PATCH	/api/orders/:orderNumber/payment	Orders (Finance)	 Done

Pending Database Work

- All models pending sequence diagrams
- Migration strategy: per-action migrations with descriptive names

Security Checklist

Feature	Status
Helmet (HTTP headers)	✅ Implemented
CORS	✅ Implemented
Rate Limiting	✅ Implemented
Cookie Security (HttpOnly, Secure)	✅ Implemented
Input Validation (Zod)	✅ Framework ready
Centralized Error Handling	✅ Implemented
Logging (Pino + redaction)	✅ Implemented
Firebase ID Token Verification	✅ Implemented (Action 1)
JWT Issuance (Kamai Access Token)	✅ Implemented (Action 1)
Refresh Token Rotation	⌚ Future action
CSRF Protection	⌚ Future action
Audit Logs	⌚ Future action
Sentry Monitoring	⌚ Pending DSN

Testing Progress

Type	Status	Coverage
Unit Tests	⌚ Pending actions	—
Integration Tests	⌚ Pending actions	—
API Tests	⌚ Pending actions	—
Coverage Threshold	Configured (70%)	—

Deployment Status

Component	Status
Dockerfile (multi-stage)	 Done
Docker Compose (API + Redis)	 Done
GitHub Actions CI	 Done
Production Environment	 Pending